Hertie School

Fall Semester 2026

Course Syllabus

GRAD-C11/ GRAD-E1339: Introduction to Data Science

1. General information

Course Format	Onsite
Instructor(s)	Prof. Simon Munzert
Instructor's E-mail	munzert@hertie-school.org
Assistant (if applicable)	Alex Karras karras@hertie-school.org
Instructor's Office Hours	Please use the online document to book an appointment

Link to Study, Examination and Admission Rules and MIA, MDS and MPP Module Handbooks

For information on **course room, times and session dates,** please consult the Course Schedule on *MyStudies*.

Instructor Information:
Simon Munzert is Professor of Data Science and Public Policy at the Hertie School and Director of the Hertie School Data Science Lab. His research interests include attitude formation in the digital age, public opinion, and the use of online data in social research. He is principal investigator in third party-funded projects on election forecasting as well as on global public opinion on hate speech regulation. He received his Doctoral Degree in Political Science from the University of Konstanz.

2. Course Contents and Learning Objectives

Course contents:
This course introduces you to the fundamental concepts of the modern data science workflow and builds the critical skills every data scientist is expected to have. Data analysis skills have become essential for those pursuing careers in policy advocacy and evaluation, business consulting and management, or academic research. Working hands-on, we will move through the full data science pipeline—from programming foundations and the command line, through data collection, wrangling, storage, and visualization, to modeling, communication, and data science ethics. Along the way we examine how AI tools are reshaping data science practice and how to use them responsibly. The course is intended for students with some prior exposure to programming.

Learning objectives:
The goals are to (1) equip you with conceptual knowledge of the data science pipeline and coding workflow, data structures, and data wrangling, (2) enable you to apply this knowledge fluently with statistical software and command-line tools, (3) help you understand the role of AI tools in the data

science workflow and how to use them responsibly, and (4) prepare you for other technical courses and the master's thesis.

Software:

We will work with RStudio/Posit, Markdown, and Git/GitHub to implement and practice the learned techniques. It is assumed that you have some basic knowledge in using R. If not, we strongly encourage you to familiarize yourself with R prior to the course so to be able to focus on the substance of the course. Good introductions (with different types of focus) are:

1. Wickham, H., Çetinkaya-Rundel M., & Grolemund, G. (2023). R for Data Science, 2nd edition. Available at: https://r4ds.hadley.nz/. This book is also the main textbook for this class, but you might want to dive in already before the course starts.
2. Ismay, Ch., & Kim, A.Y. (2023). Statistical Inference via Data Science. A Modern Dive into R and the Tidyverse. Available at: https://moderndive.com/
3. DataQuest (https://www.dataquest.io/), in particular the following courses from the "Data Analyst in R" career path:
    a. Introduction to Data Analysis in R
    b. Data Structures in R
    c. Control Flow, Iteration, and Functions in R
    d. Specialized Data Procession in R
    e. Data Cleaning in R

Target group:
MDS 1st year students; MPP/MIA 2nd year students who have successfully completed Statistics II.

Teaching style:
The sessions will feature a lecture on the session's topic led by the instructor. For the discussion of exercises and their implementation in R, students will be enrolled in a small group support lab session taught weekly by a teaching assistant.

Prerequisites:
For MPP/MIA 2nd year students: In order to take this course as an elective, it is required that you have successfully completed Statistics II. (Taking Statistics II in parallel is not sufficient.)

Diversity Statement**:**
Understanding and respect for all cultures and ethnicities is central to the teaching at Hertie. Being mindful of diversity is an important issue for policy professionals in the planning, implementation, and evaluation of programmes designed for specific groups, populations, or communities. Diversity and cultural awareness will be integrated in the course content whenever possible.

3. Grading and Assignments

Policy on the use of AI tools for assignments

The use of generative AI tools (such as ChatGPT) is permitted only for specific assignments. Students must adhere to the specific instructions provided for each assignment. Where AI tools are used, students are expected to clearly indicate how they were used and to ensure the originality and academic integrity of their work. **Students must include a Statement of Authorship in all written assignments.** Improper or undisclosed use of AI tools may be considered a violation of academic integrity policies. Please consult the course instructor in case of doubt. Further guidance is available in the **[Artificial Intelligence Tools at the Hertie School - Teaching Guidelines for Instructors and Students](#)**, including a template for the **Statement of Authorship, under section 1.1.**

Composition of Final Grade:

Assignment 1: Homework assignments (H)	Thursdays in the week after the lecture at 10am	Submit via GitHub	pass/fail
Assignment 2: Quizzes (Q)	At the beginning of the lecture	Onsite	8(10) x 3% = 24%
Assignment 3: Hackathon project	72h sprint (start: November 6, 12pm, end: November 9, 12pm)	Submit via GitHub	36%
Assignment 4: In-class exam	Final exam week, exact date TBD	Onsite	40%

Assignment Details

Evaluation is conducted via a combination of a series of homework assignments (pass/fail; not contributing to the final grade), quizzes (counting towards 24% of the final grade), a hackathon project (counting towards 36% of the final grade), and an in-class exam (counting towards 40% of the final grade). While you should submit your own, individual solutions to the homework assignments, I generally encourage you to study and learn to use the software together.

Assignment 1: Homework assignments
In a series of homework assignments, you will apply the concepts learned in class to solve data analytic problem sets using R. While you are encouraged to collaborate, everyone will hand in a separate solution. Submissions are assessed on a pass/fail basis. You must pass at least 75% of those assignments.

Assignment 2: Quizzes
In 10 short quizzes, administered at the beginning of the lecture, you will be tested on the concepts learned in class. The 8 quizzes will contribute to the final grade.

Assignment 3: Hackathon project
In the hackathon project, you get to solve a set of problems related using one given dataset and also submit a creative project of your choice using the same dataset. After the instructions have been made available, you will have 72 hours to submit solutions as a team of three to four people. The task is similar to the homework assignments but puts more emphasis on creative problem-solving using the tools and techniques you have learned in class. The hackathon runs as an extracurricular activity in the week before the communication and big picture block. **The tentative time slot for the hackathon kick-off is November 6, 10am-12pm.**

Assignment 4: In-class exam

You will complete an exam during class time on pen and paper. No laptops, internet access, or other electronic aids are permitted. Communication or collaboration with other students—either in person or online—is strictly prohibited. The exam is designed to assess individual problem-solving skills and coding proficiency under timed conditions.

Late submission of assignments: For each day the assignment is turned in late, the grade will be reduced by 10% (e.g. submission two days after the deadline would result in 20% grade deduction).

Attendance: Students are expected to be present and prepared for each class session. Active participation during lectures and seminar discussions is essential. Please note that students can miss up to two sessions (out of twelve) if no course assignments are affected. **Students do not need to inform the course instructor(s) in advance if they are unable to attend a session. In fact, I (the instructor) don't want you to inform me.** For further information please consult the Examination Rules §10.

Academic Integrity: The Hertie School is committed to the standards of good academic and ethical conduct. Any violation of these standards shall be subject to disciplinary action. Plagiarism, misuse of AI, free riding in group work, and other deceitful actions are not tolerated. See Examination Rules §16, the Hertie Plagiarism Policy, and the Hertie Guidelines for Artificial Intelligence Tools.

Compensation for Disadvantages: If a student furnishes evidence that they are not able to take an examination as required in whole or in part due to disability or permanent illness, the Examination Committee may upon written request approve learning accommodation(s). In this respect, the submission of adequate certificates may be required. See Examination Rules §14.

Extenuating circumstances: An extension can be granted due to extenuating circumstances (i.e., for reasons like illness, personal loss or hardship, or caring duties). In such cases, please contact the course instructor and Examination Office *in advance* of the assignment deadline.

4. Course Sessions and Readings

All course readings can be accessed on the course Moodle page.

Please note that fall semester Moodle course pages are available until 31 March of the following semester. Spring semester Moodle courses are available until 31 September of the following semester. After that, in accordance with German copyright law, they cannot be accessed.

Session	Session Title	Date
	Fundamentals	
1	**What is data science?**	**September 03**
2	**Programming I: Functions and debugging**	**September 10**
3	**Programming II: Iteration, automation, scheduling**	**September 17**
4	**Working with the command line**	**September 24**
	Collecting data	
5	**Web data and technologies**	**October 01**
6	**Web scraping and APIs**	**October 08**

7	**Relational databases and SQL**	**October 15**
<u>Fall break: no classes</u>		
<u>Analyzing data</u>		
8	**Modeling**	**October 29**
9	**Visualization**	**November 05**
–	**Hackathon (extracurricular)**	**Extracurricular (week of Nov 09)**
<u>Recurring parts of the workflow</u>		
10	**Monitoring and communication**	**November 12**
11	**Data science ethics**	**November 19**
12	**AI as part of the data science workflow**	**November 26**
<u>Make up week and Final exam week – no classes</u>		

If not freely available online (see URLs), all readings will be accessible on the Moodle course site before semester start. In the case that there is a change in readings, students will be notified by email.

Required readings are to be read and analyzed thoroughly. Optional readings are intended to broaden your knowledge in the respective area, and it is highly recommended to at least skim them.

During this course, we will frequently rely on the following textbook:

> Wickham, H., Çetinkaya-Rundel M., & Grolemund, G. (2023). *R for Data Science, 2nd edition*. O'Reilly. Free online version available at: https://r4ds.hadley.nz/ [**R4DS**]

If you already have substantial experience with R, you're recommended to also check out the following textbook, which will be considered optional reading in some sessions:

> Wickham, H., (2023). *Advanced R, 2nd edition*. O'Reilly. Free online version available at: https://adv-r.hadley.nz/ [**AdvR**]

Session 1: What is data science?	
Learning Objective	After this session you have an overview of data science as an interdisciplinary field. Also, you'll have a roadmap of the course to navigate the rest of the course. In the accompanying lab, you will be geared up to wrangle data in R the "modern", tidyverse way.
Required Readings	1. Kelleher, John and Brendan Tierney. 2018. *Data Science.* MIT Press. Chapter 1: What Is Data Science? 2. Salganik, Matt. 2017. *Bit by Bit: Social Research in the Digital Age.* Princeton University Press. Chapters 1 and 2.
Optional Readings	3. **R4DS**. Chapters 1—4. 4. **AdvR.** Chapters 1—5.

Session 2: Programming I: Functions and debugging	
Learning Objective	After this session, you (a) can make your code more efficient using functional programming tools, and (b) have learned the basics of debugging, starting with a general strategy, then following up with specific tools. Plus, we'll discuss a robust project setup.
Required Readings	1. **R4DS**. Chapter 25. 2. **AdvR.** Chapter 22.
Optional Readings	3. **AdvR.** Chapters 6-8. 4. Grolemund, G. Hands-On Programming with R. https://rstudio-education.github.io/hopr/.

Session 3: Programming II: Iteration, automation, scheduling	
Learning Objective	We will continue our treatment of functional programming – this time it is all about repetitive tasks and how to tackle them using tools for iteration, automation, and scheduling.
Required Readings	1. **R4DS**. Chapter 26. 2. Bryan, Jenny. 2019. All the automation things. https://stat545.com/automation-overview.html
Optional Readings	3. **AdvR.** Chapter 9-11. 4. Wickham, Hadley. 2015. *R Packages, 2nd edition*. O'Reilly. Chapters 1—5. https://r-pkgs.org/

Session 4: Working with the command line	
Learning Objective	After this session, you (a) are comfortable navigating the file system and running programs from the shell, (b) can chain basic commands and pipes to manipulate files and text, and (c) understand how the command line fits into a reproducible data science workflow.
Required Readings	The Unix Workbench (Sean Kross). Available at: https://seankross.com/the-unix-workbench/ Chapters on the command line and shell scripting.
Optional Readings	Software Carpentry. The Unix Shell. Available at: https://swcarpentry.github.io/shell-novice/ Janssens, Jeroen. Data Science at the Command Line. *2nd edition*. O'Reilly. Available at: https://jeroenjanssens.com/dsatcl/ Command line learning game Terminus: https://web.mit.edu/mprat/Public/web/Terminus/Web/main.html

Session 5 Web data and technologies	
Learning Objective	After this session, you have acquired basic knowledge of web technologies and web data in its various forms.
Required Readings	1. **R4DS.** Chapters 12—16; with focus on 15. 2. Munzert, Simon, Christian Rubba, Peter Meißner, and Dominic Nyhuis. 2015. *Automated Data Collection with R. A Practical Guide to Web Scraping and Text Mining*. Chichester: John Wiley & Sons. Chapters 1 (Introduction), 2 (HTML), 3 (XML and JSON), 4 (XPath), 8 (Regular expressions)
Optional Readings	3. Regexcrossword – the regular expressions game. https://regexcrossword.com/ 4. https://cran.r-project.org/web/views/WebTechnologies.html 5. https://github.com/jeroen/jsonlite

Session 6: Web scraping and APIs	
Learning Objective	After this session, you (a) have acquired basic knowledge of classic web scraping and (b) web APIs.
Required Readings	1. **R4DS**. Chapter 24. 2. Bauer, Paul, Lion Behrens, and Camille Landesvatter (eds.). 2022. *APIs for social scientists: A collaborative review.* https://paulcbauer.github.io/apis_for_social_scientists_a_review/ Chapters 1 (Introduction), 2 (Best Practices), and any of the API-specific chapters you find interesting (at least one to get an impression!)
Optional Readings	1. Munzert, Simon, Christian Rubba, Peter Meißner, and Dominic Nyhuis, 2015: *Automated Data Collection with R. A Practical Guide to Web Scraping and Text Mining*. Chichester: John Wiley & Sons. Chapter 9 (Scraping the web)

Session 7: Relational databases and SQL	
Learning Objective	After this session you will (a) understand the principles of relational data structures and basic normalization, (b) be able to manipulate and join tables of data, (c) be able to interact with remote relational databases as if local.
Required Readings	1. **R4DS**. Chapters 19—21.
Optional Readings	2. Munzert, Simon, Christian Rubba, Peter Meißner, and Dominic Nyhuis, 2015: *Automated Data Collection with R. A Practical Guide to Web Scraping and Text Mining*. Chichester: John Wiley & Sons. Chapter 7 (SQL and relational databases) 3. Weidmann, Nils. 2023. *Data Management for Social Scientists. From Files to Databases.* Cambridge University Press.

	4. Lord, Robin. Lost at SQL – the learning game. https://lost-at-sql.therobinlord.com/

Session 8: Modeling	
Learning Objective	After this session you will learned the basics of model fitting and evaluation techniques with R.
Required Reading	1. https://www.tidymodels.org/start/, Get Started, Articles 1-5 2. Simonsohn, Uri, Joseph P. Simmons, Leif D. Nelson. 2020. Specification curve analysis. *Nature Human Behaviour* 4, 1208—1214.
Optional Readings	3. Dani Cosme. 2022. Specification Curve Analysis (tutorial) https://dcosme.github.io/specification-curves/SCA_tutorial_2022

Session 9: Visualization	
Learning Objective	After this session, you (a) have learned about basic rules to making visualizations that accurately reflect the data, tell a story, and look professional, (b) have learned about popular mistakes in visualization and how to avoid them, and (c) are able to integrate visualization as complementary means to analyze data into your workflow.
Required Readings	1. Wilke, Claus. 2019. *Fundamentals of data visualization: a primer on making informative and compelling figures*. O'Reilly Media. https://clauswilke.com/dataviz/. 2. **R4DS**, Chapters 9—11.
Optional Readings	3. Healy, Kieran. 2018. *Data visualization: a practical introduction*. Princeton University Press. https://socviz.co/ 4. Traunmüller, Richard. 2020. Visualizing Data in Political Science. In: L. Curini & R. Franzese (eds.) *The SAGE Handbook of Research Methods in Political Science and International Relations*. Sage. https://tinyurl.com/visualization-polisci

Session 10: Monitoring and communication	
Learning Objective	After this session, you will learn how to (a) build interactive web apps and dashboards using Quarto/Markdown and Shiny, and (b) communicate your results and products.
Required Readings	1. **R4DS**. Chapters 28—29. 2. https://shiny.rstudio.com/
Optional Readings	3. van der Bles AM, van der Linden S, Freeman ALJ, Mitchell J, Galvao AB, Zaval L, Spiegelhalter DJ. 2019 Communicating uncertainty about facts, numbers and science. *R. Soc. open sci.* 6: 181870. http://dx.doi.org/10.1098/rsos.181870

	4. Wickham, Hadley. 2020. *Mastering Shiny*. O'Reilly. https://mastering-shiny.org/

Session 11: Data science ethics	
Learning Objective	After this session, you will have a deeper understanding of the ethical issues associated with working with sensitive data or with the construction of tools that - if applied improperly - can do harm to research subjects or society as a whole.
Required Readings	1. Salganik, Matt. 2017. *Bit by Bit: Social Research in the Digital Age.* Princeton University Press. Chapter 6 (Ethics) 2. Kelleher, John and Brendan Tierney. 2018. *Data Science*. MIT Press. Chapter 6 (Privacy and Ethics)
Optional Readings	3. Floridi, Luciano and Mariarosaria Taddeo. 2016. What is Data Ethics? *Phil. Trans. R. Soc. A. 37420160360.*

Session 12: AI as part of the data science workflow	
Learning Objective	After this session, you will be able to (a) reflect critically on where AI tools fit into the data science workflow, (b) use them responsibly and transparently as an aid to rather than a substitute for your own coding and reasoning, and (c) recognize their limitations and the risks of over-reliance.
Required Readings	1. Spirling, A. 2023. Why open-source generative AI models are an ethical way forward for science. *Nature* 616, 413. 2. Cheng, Joe, Rohrbaugh, N., and Altman, S. 2026. Introducing AI in RStudio. https://posit.co/blog/introducing-ai-in-rstudio.